PathwayAI | Orange County

# Lyme Disease Burden & Community Action

Prepared 2026-11-02 · Public-data planning brief · Version 1.8

For local planning discussions · Proposed actions for review

| Reported cases · 2024 | County rate per 100,000 · 2024 | State rate per 100,000 · 2024 |
|---|---|---|
| 964 | 237 | 109.8 |

## Illustrative economic burden estimate

# USD 2.63 million

Optional 2025 purchasing-power comparison · 11 available CPI months · general inflation

| Original study basis · 2016 dollars | Optional 2025 comparison · approximately 1.34× |
|---|---|
| USD 1.96 million | USD 2.63 million |

Study-based illustration, not measured county spending or predicted savings. Sources [1, 5, 8]; calculation and methods: page 3.

## Mid-Hudson counties and state comparison

| Area | Reported cases · 2024 | Rate per 100,000 · 2024 |
|---|---|---|
| Dutchess | 1,060 | 355.5 |
| Orange | 964 | 237 |
| Putnam | 294 | 299.2 |
| Rockland | 461 | 135.3 |
| Sullivan | 385 | 482.6 |
| Ulster | 841 | 461.2 |
| Westchester | 1,125 | 113.2 |
| New York State | 21,632 | 109.8 |

All comparison rates use the same NYSDOH 2024 published source and reporting year; reported rates are not personal risk estimates.

## Healthcare resources and population context

| Indicator | Value | Period |
|---|---|---|
| Primary-care physicians | 294 | 2023 |
| Hospitals | 6 | 2023 |
| Hospital beds | 1,111 | 2023 |
| Uninsured residents under age 65 | 5.5% | 2022 |
| Poverty (population with poverty status determined) | 12.7% | 2023 |
| Language other than English at home · age 5+ | 30.2% | ACS 2020–2024 |

General resource/population context, not Lyme-attributable need, appointment availability or interpreter demand. Selected insurance/poverty fields supply no margin of error.

## Reported Lyme rates by year · per 100,000 residents

| 2022 | 2023 | 2024 |
|---|---|---|
| 231.6 | 296.9 | 237 |

County of residence, not exposure location. Pre-2022 rates excluded because surveillance definitions changed. Different sources do not establish causes. References and limitations: page 3.

# From evidence to community action

## Published components · per participant · 2016 dollars

Medical: USD 1,333; other expenses and productivity combined: USD 699; total: USD 2,032.

The combined component is a derived residual, not a separately reported mean. A separate caregiving amount is not established here; do not add components to the total again.

## Support priority: Getting to care

Question: Should a county care-navigation team investigate a small ride-navigation pilot with willing residents?

Possible response: Ask willing residents and transport providers to review ride eligibility, accessibility and appointment coordination.

Who could help: Patient and caregiver representatives, transport services and care navigators

## Actions to discuss · Public health departments

- Share tick-prevention information through local events and trusted community channels.
- Review county surveillance alongside service availability; ask residents which barriers need attention.
- Bring residents and potential partners together to choose one support priority and agree how to follow up.

## Four areas for community action

Prevention education and tick-bite awareness: Share established prevention information through trusted local channels.

Timely access to healthcare: Help residents find care and ask about appointments, costs, transport and interpreters.

Financial and caregiving burden: Recognize spending, missed work, unpaid care and support needs beyond medical bills.

Language access and community resources: Make information easier to understand and connect residents with existing local support.

## Resident priorities to discuss

Confirm priorities with residents and caregivers before choosing a response. No county-specific patient-reported findings are established in this brief.

## Before starting a pilot

- Confirm local demand with residents and caregivers, including assisted/offline participation.
- Verify current services, eligibility, interpreter access and appointment capacity.
- Agree who is responsible and estimate staff time, program costs and a funding source.
- Define consent, minimal data collection, access, retention and deletion before any collection.

## Check whether people received support

Choose a common follow-up period, such as 30 days. Record the starting situation, help received and needs still unmet. Keep unanswered questions unknown.

For this priority: Requested and completed accessible rides; appointments attended; travel spending; support still needed

Further evaluation is needed to establish health benefits or savings. Actions are proposals for local review, not adopted programs or commitments.

## Professional review

Public-health, health-economics and Lyme-care reviewers: please comment on source clarity, cost assumptions, practical usefulness and missing local factors.

Email feedback: pathwayai.feedback@gmail.com

# References & methods

## Original study basis

964 reported cases × USD 2,032 study mean = USD 1,958,848 (2016 dollars).

Hook et al. (published 2022): 613 participants in societal analysis; reported cases recruited in four high-incidence states during 2014–2016. Societal: medical, nonmedical and productivity costs; components are already included.

Applying this study mean to reported county cases is an assumption; the result is not measured local spending. No adjustment for underreporting or local case mix. No causal savings or disability probabilities are inferred.

Hook excluded non-English speakers and nonacute disease. The study does not represent every local household or continuing lifetime costs.

## Optional inflation comparison

General CPI factor approximately 1.34; calculations use the full-precision factor retained in the methods CSV. Adjusted study mean approximately USD 2,726 per participant (rounded for display).

2016 uses 12 monthly observations; 2025 uses 11 available months. October is missing; no imputation. This is not a verified official full-year annual average or medical-care inflation adjustment. It changes purchasing power, not case mix, treatment patterns or local prices.

## Source register


1. Reported cases, county of residence · 2024

https://www.health.ny.gov/statistics/diseases/communicable/2024/docs/cases.pdf

2. Reported rates per 100,000 · 2024

https://www.health.ny.gov/statistics/diseases/communicable/2024/docs/rates.pdf

3. CDC surveillance interpretation; historical county file in app · 2019–2022 county file; separate from NYSDOH 2024

https://www.cdc.gov/lyme/data-research/facts-stats/index.html

4. Healthcare resources, insurance and poverty context · 2022 insurance fields; 2023 healthcare and poverty fields; AHRF 2024–2025 release

https://data.hrsa.gov/topics/health-workforce/nchwa/ahrf

5. Societal study mean · 2016 dollars; recruitment 2014–2016

https://wwwnc.cdc.gov/eid/article/28/6/21-1335_article

6. Shared Mid-Hudson regional assessment (Dutchess and Orange), Figure 147, page 219 · 2022–2023; regional assessment published 2025

https://www.dutchessny.gov/Departments/DBCH/Docs/MHRCHA2025.pdf

7. Language spoken at home, age 5+ · ACS 2020–2024

https://www.census.gov/quickfacts/orangecountynewyork

8. BLS CPI-U via supplied FRED file · 2016 and 2025 available-month comparison

https://fred.stlouisfed.org/series/CPIAUCNS


## Verification and interpretation

Existing source snapshots; newer official releases have not been verified for this brief. Official rates are reproduced; exact rate-denominator vintage is not independently verified. Pre-2022 rates are excluded because surveillance definitions changed.

Census language context is an existing QuickFacts snapshot, not a fresh pull or a measure of English proficiency. HRSA field years are not substituted with newer Census periods. Initiative links require current local verification; no listing does not mean no services exist.

Public evidence and proposed planning actions only. No session stories, personal records or synthetic patient cards are included.

Discussion brief, not a county demand assessment, benefits decision, funded service or adopted county plan. No county or organizational endorsement is implied.

# What published research adds

## Research context · external study populations

These studies help explain burdens worth considering. They are not county patient findings, and their estimates are not added together or applied as county disability rates.

### Hook et al. (2022)

901 patient-cost participants; 613 societal-cost participants · Recruitment 2014–2016; 2016 dollars

Medical, nonmedical and productivity costs inform the existing illustration.

Population and limits: Selected reported cases in four high-incidence states; excluded non-English speakers and nonacute disease.

https://wwwnc.cdc.gov/eid/article/28/6/21-1335_article

### Adrion et al. (2015)

52,795 Lyme patients; 263,975 matched controls · Claims 2006–2010; 12-month follow-up

Documents healthcare spending and utilization after Lyme diagnosis.

Population and limits: Commercially insured ages 0–64; not uninsured patients or a full household-cost estimate.

https://doi.org/10.1371/journal.pone.0116767

### Johnson et al. (2014)

3,090 selected survey respondents · Survey 2013; persistent symptoms after treatment

Documents reported work disruption, expenses and quality-of-life impacts.

Population and limits: Self-selected, self-reported persistent-illness sample; not all Lyme cases or a county disability rate.

https://doi.org/10.7717/peerj.322

## What this means for local planning

Prevention outreach: Orange's 2024 reported rate was 237 per 100,000, above the state rate of 109.8. Where could trusted local channels help share prevention information?

NYSDOH 2024

Care access: 5.5% of residents under age 65 were uninsured in 2022. Could care navigators help residents ask about sliding fees and coverage options? This is general population context, not Lyme-specific need or available appointments.

HRSA AHRF · insurance 2022

A question for local review: Should a county care-navigation team investigate a small ride-navigation pilot with willing residents? Confirm needs with residents and check existing services before selecting a response.

Suggested planning question, not a local finding

## How to use this evidence

Use these studies to discuss spending, work disruption and daily-life impacts with local residents. Hook remains the sole study-cost basis for the county illustration. No local disability rate, causal savings or added cost total is inferred.

The project survey is excluded from this county brief. County-specific patient findings would require a defined local collection and review process.